# Kim 2018 TNBC — analysis summary

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: same-sample neighbour fraction 0.99→0.40; iLISI 3.89 of 9 samples; continuity confirmed off the independent PCA-harmony graph (1 connected component), 0 islands.

### Gate 2 — module geography → GO (non-branching)

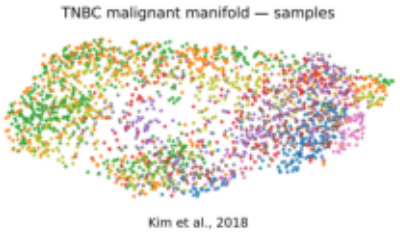
Single axis: pSom-high origin → Gast/Crest-high terminus. Organized Gast↔pSom cross-repression (smoothed r=−0.84); Crest a terminal gradient. FLAG: Crest enriches the Gast pole (inversion) — passes anyway.

### Gate 3 — cross-repression → NA

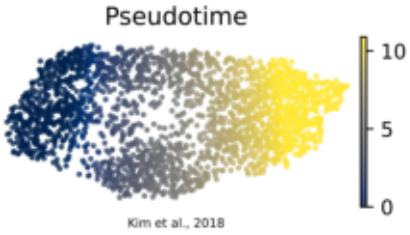
No bifurcation (curve: 2 tips, 0 forks) — branching-only Gate 3 not applicable.

## STEP FIGURES

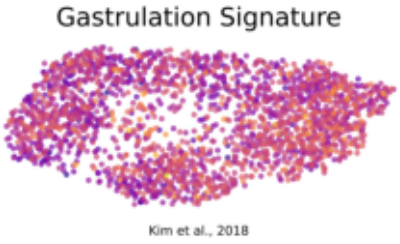
**Manifold — samples**



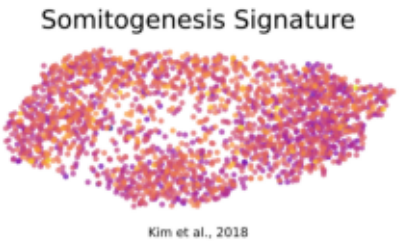

**Pseudotime**



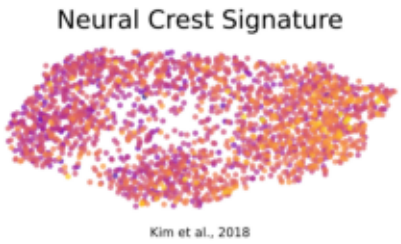

**Gastrulation**



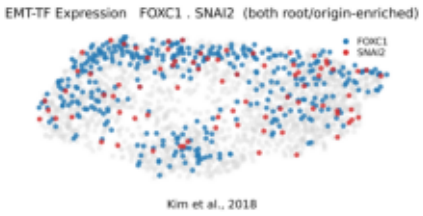

**Somitogenesis**




**Neural Crest**




**EMT-TF expression**




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**Dataset.**
Triple-negative breast cancer (Kim et al., 2018), Nanogrid — 2,263 malignant cells across 9 samples (9/16 kept at ≥100-cell floor). Non-branching Linear-pSom/Crest.

**Verdict.**
The embryonic-program pattern reproduces as a non-branching linear axis: a single continuous, sample-integrated manifold from a pSom-high developmental origin to a Gast/Crest-high terminus, with organized Gast↔pSom cross-repression and Crest as a terminal gradient enriched on the Gast pole (inversion flag). No bifurcation → Gate 3 NA. EMT-TF expression (FOXC1, SNAI2) concentrates at the root/origin end; no EMT-TF marks the terminus in this dataset.

## Parameter choices

| | | | |
|---|---|---|---|
| **Min-cell threshold** | ≥100 (floor coincides with real 52→100 inflection) → 9/16, 2,263 cells | **Embedding** | HVG=536; PCA50; Harmony θ=10 (TNBC patient batch); n_pcs=20, spread=1.3 |
| **Trajectory** | scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.98) | **Root** | pSom-high/Crest-low developmental origin (Crest is late) |
| **Geometry** | Linear-pSom/Crest (raw Gast −0.009, pSom +0.040, Crest −0.085) | **EMT-TFs** | FOXC1 (65%, ρ=−0.41) + SNAI2 (ρ=−0.23); BOTH root/origin-enriched, none terminal |